MSG No. 176.146 $P6_3/m'$ [Type III, hexagonal]

Table 1: Wyckoff site: `2a`, site symmetry: `-6'..`

No.	position	mapping
1	$\left[0,\ 0,\ \frac{1}{4}\right]$	`[1,3,5,8,10,12]`
2	$\left[0,\ 0,\ \frac{3}{4}\right]$	`[2,4,6,7,9,11]`

Table 2: Wyckoff site: `2b`, site symmetry: `-3'..`

No.	position	mapping
1	$[0,\ 0,\ 0]$	`[1,3,5,7,9,11]`
2	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[2,4,6,8,10,12]`

Table 3: Wyckoff site: `2c`, site symmetry: `-6'..`

No.	position	mapping
1	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{4}\right]$	`[1,3,5,8,10,12]`
2	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{3}{4}\right]$	`[2,4,6,7,9,11]`

Table 4: Wyckoff site: `2d`, site symmetry: `-6'..`

No.	position	mapping
1	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{4}\right]$	`[1,3,5,8,10,12]`
2	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{3}{4}\right]$	`[2,4,6,7,9,11]`

Table 5: Wyckoff site: `4e`, site symmetry: `3..`

No.	position	mapping
1	$[0,\ 0,\ z]$	`[1,3,5]`
2	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	`[2,4,6]`
3	$[0,\ 0,\ -z]$	`[7,9,11]`
4	$\left[0,\ 0,\ \frac{1}{2}-z\right]$	`[8,10,12]`

Table 6: Wyckoff site: `4f`, site symmetry: `3..`

No.	position	mapping
1	$\left[\frac{1}{3},\ \frac{2}{3},\ z\right]$	`[1,3,5]`
2	$\left[\frac{2}{3},\ \frac{1}{3},\ z+\frac{1}{2}\right]$	`[2,4,6]`
3	$\left[\frac{2}{3},\ \frac{1}{3},\ -z\right]$	`[7,9,11]`
4	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{2}-z\right]$	`[8,10,12]`

Table 7: Wyckoff site: `6g`, site symmetry: `-1'`

No.	position	mapping
1	$\left[\frac{1}{2},\, 0,\, 0\right]$	`[1,7]`
2	$\left[\frac{1}{2},\, \frac{1}{2},\, \frac{1}{2}\right]$	`[2,8]`
3	$\left[0,\, \frac{1}{2},\, 0\right]$	`[3,9]`
4	$\left[\frac{1}{2},\, 0,\, \frac{1}{2}\right]$	`[4,10]`
5	$\left[\frac{1}{2},\, \frac{1}{2},\, 0\right]$	`[5,11]`
6	$\left[0,\, \frac{1}{2},\, \frac{1}{2}\right]$	`[6,12]`

Table 8: Wyckoff site: `6h`, site symmetry: `m'..`

No.	position	mapping
1	$\left[x,\, y,\, \frac{1}{4}\right]$	`[1,10]`
2	$\left[x-y,\, x,\, \frac{3}{4}\right]$	`[2,11]`
3	$\left[-y,\, x-y,\, \frac{1}{4}\right]$	`[3,12]`
4	$\left[-x,\, -y,\, \frac{3}{4}\right]$	`[4,7]`
5	$\left[-x+y,\, -x,\, \frac{1}{4}\right]$	`[5,8]`
6	$\left[y,\, -x+y,\, \frac{3}{4}\right]$	`[6,9]`

Table 9: Wyckoff site: `12i`, site symmetry: `1`

No.	position	mapping
1	$[x,\, y,\, z]$	`[1]`
2	$\left[x-y,\, x,\, z+\frac{1}{2}\right]$	`[2]`
3	$[-y,\, x-y,\, z]$	`[3]`
4	$\left[-x,\, -y,\, z+\frac{1}{2}\right]$	`[4]`
5	$[-x+y,\, -x,\, z]$	`[5]`
6	$\left[y,\, -x+y,\, z+\frac{1}{2}\right]$	`[6]`
7	$[-x,\, -y,\, -z]$	`[7]`
8	$\left[-x+y,\, -x,\, \frac{1}{2}-z\right]$	`[8]`
9	$[y,\, -x+y,\, -z]$	`[9]`
10	$\left[x,\, y,\, \frac{1}{2}-z\right]$	`[10]`
11	$[x-y,\, x,\, -z]$	`[11]`
12	$\left[-y,\, x-y,\, \frac{1}{2}-z\right]$	`[12]`